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MAI(A) the Sound Be With You

MAIA-AV: A Ground-Up Audio-Visual Benchmark for Multimodal AI Assessment

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Introduction

The rapid evolution of Vision-Language Models raises the need to determine whether their strong performance reflects genuine multimodal understanding or the exploitation of superficial correlations and linguistic biases. In this contex *MAIA: Multimodal AI Assessment* by Testa et al.(2025)[21, 20] was introduced as a native Italian benchmark for evaluating video-based reasoning in Vision-Language Models. Testa et al. investigate the extent to which VLMs coherently integrate visual and linguistic information. More specifically, it assesses whether model predictions are grounded in visual evidence rather than primarily driven by distributional regularities learned by the language component. This distinction is crucial because a model may answer a question correctly while failing to recognise the same information consistently when it is presented in a discriminative format. In such cases, the prediction may result from linguistic plausibility rather than genuine visual understanding. To address this issue, MAIA employs two aligned tasks: Visual Statement Verification and Open-Ended Visual Question Answering. Through these tasks, Testa et al. evaluate not only models accuracy, but also the robustness and consistency of models behaviour across different evaluation formats. The benchmark further defines twelve fine-grained reasoning categories to identify which competencies are used by the models and in which areas linguistic shortcuts, unimodal collapse, or hallucinations emerge. A limitation of the original framework is that the dataset was designed and evaluated primarily on the basis of visual information. Real-world videos, however, are inherently audiovisual: relevant meaning may also be conveyed through dialogue, environmental sounds, music, and other acoustic events. Recent multimodal and omnimodal architectures can process visual, auditory, and textual inputs, but the availability of multiple modalities does not guarantee that they will be integrated effectively. Models may over-rely on a single channel or be negatively affected by redundant or weakly relevant information. Starting from this problem, the study Deodati,Kaif et al.(2026): "*Lost in Transcription: Investigating the Role of Audio Information for Video-based Visual Statement Verification*"[5] extends the original MAIA research framework to a more complex analysis of the interaction between visual, linguistic, and auditory information, while preserving its competence-oriented perspective. The study considers two complementary representations of audio: raw signal and automatic transcriptions provided as textual input. By evaluating these representations both independently and in combination with video information, the analysis investigates whether audio contributes meaningfully to task performance and whether one representation has a stronger influence on model predictions. This leads to two research questions:

RQ1: *To what extent does raw audio affect video-based Visual Statement Verification when it is jointly processed with visual and textual information?*

RQ2: *How effectively do multimodal models integrate visual, textual, and auditory information when performing video-based reasoning and understanding tasks?*

The experiments show that raw audio has a limited and strongly context-dependent effect. It is most useful when visual information is unavailable, whereas, when added to an already informative visual input, it may provide little benefit or introduce noise. The results also show that transcriptions can produce greater interference than raw audio, suggesting that multimodal architectures do not necessarily achieve more effective cross-modal understanding simply because additional modalities are available. These findings must nevertheless be interpreted in relation to the nature of the dataset. The one hundred original MAIA videos were selected and annotated according to their visual content, and the questions were formulated without considering the audio track. Consequently, the benchmark does not systematically require auditory information to solve its items. My thesis project begins from this limitation. The following chapters investigate whether models can integrate multiple input modalities more coherently when the dataset is specifically designed so that auditory information is at least partially relevant. The central hypothesis is that, under these conditions, models may use complementary signals across modalities to reconstruct missing information and develop a more genuinely multimodal understanding of video content, producing answers grounded in the available evidence rather than in linguistic plausibility or chance.

1. Literature review

The rapid evolution of Visual Language Models (VLMs) has progressively expanded the ability of neural systems to jointly process perceptual and linguistic information. This line of research builds on the hypothesis that linguistic meaning cannot be constructed exclusively from distributional regularities learned from textual corpora, but must also be grounded in perceptual experience [2]. Within this context, Visual Question Answering (VQA) emerged as one of the first systematic paradigms for assessing whether a model can answer questions about the content of an image [1]. Subsequent benchmarks, such as GQA by Hudson and Manning (2019), extended this setting by introducing compositional questions involving entities, attributes, and visual relations [9]. However, high task accuracy alone does not provide sufficient evidence of multimodal comprehension. Several studies have shown that models can exploit statistical regularities in answers, questions, or dataset distributions, producing correct predictions without necessarily *understanding* the associated visual content. Evaluation has therefore progressively moved beyond task performance towards the assessment of grounding, understood as the ability to associate a prediction with the relevant perceptual evidence. One widely adopted strategy for evaluating grounding relies on minimally contrastive pairs, in which a correct instance is paired with a foil obtained by modifying a semantically relevant element. *FOIL-it!*, for example, evaluates whether a model can detect localized inconsistencies between an image and its linguistic description [18]. *Winoground*, instead, presents pairs of images and descriptions containing the same lexical elements but arranged according to different compositional relations, showing that recognizing individual entities does not necessarily entail understanding the relational structure connecting them [22]. The transition from images to videos introduces additional challenges. Video understanding requires integrating information distributed over time, recognizing actions and state changes, ordering events, and establishing temporal and causal relations. *Action Genome Question Answering (AGQA)* addresses these requirements through compositional questions concerning objects, actions, and spatio-temporal relations [8]. Similarly, the *Multi-modal*

Video Understanding Benchmark (MVBench) evaluates a broader range of competencies, including action perception, movement, temporal ordering, and dynamic reasoning [13]. Nevertheless, video benchmarks remain vulnerable to unimodal shortcuts and to strategies that rely predominantly on the most informative frame. Kesen et al. (2024) address this limitation with *Video Language Model Assessment (VILMA)*, a zero-shot benchmark based on video–text contrastive pairs [10]. For each complex capability, VILMA introduces a proficiency test designed to verify the preliminary perceptual ability required to solve the corresponding task. The results show that, once these prerequisite conditions are considered, part of the apparently correct performance can be attributed to accidental or spurious strategies. In particular, the examined VLMs do not display a systematic advantage over statistical image-based models on tasks requiring strict temporal grounding. Within this line of research, MAIA introduced a natively Italian benchmark designed to provide a fine-grained evaluation of video-linguistic reasoning [21]. The resource covers twelve reasoning categories related to linguistic phenomena, including causal, counterfactual, implicit, spatial, temporal, sentimental, planning, uncertainty, and out-of-scope reasoning. Its main contribution lies in aligning two tasks built on the same videos and questions: *Visual Statement Verification (VSV)*, where the model selects the correct statement from a minimally contrastive pair, and *Open-Ended Visual Question Answering (OEVQA)*, where the model freely generates the answer. This setting makes it possible to assess not only accuracy but also coherence between discriminative comprehension and linguistic generation. A model may correctly answer a multiple-choice question while failing to reproduce the same information in an open-ended setting, or may behave inconsistently across equivalent linguistic formulations. To capture this phenomenon, MAIA introduces the *Aggregate Accuracy* metric, which rewards only those instances in which the model correctly answers all true–false pairs associated with the same question and simultaneously generates the correct response in the corresponding open-ended task. The reduction in Aggregate Accuracy relative to conventional single-instance accuracy highlights the fragility of apparently strong performance and the need to jointly evaluate robustness, consistency, and grounding. Although VLMs have traditionally focused on text–vision interactions, <u>real videos are intrinsically audiovisual</u>. Their interpretation may depend on music, prosody, background noise, and other acoustic signals that cannot be reconstructed from visual frames alone. Consequently, an increasing body of research has extended multimodal language model architectures to support joint processing of text, images, video, and audio. *ImageBind*, for instance, introduced a shared representation space spanning images, text, audio, depth, thermal information, and inertial movement [7]. Other multimodal language models (MLMs) rely on modality-specific encoders combined with projection or fusion modules that align heterogeneous representations with the linguistic space. *Video-SALMONN: Speech-Enhanced Audio-Visual Large Language Models* introduces a multi-resolution Q-former[1] to align audio signals with synchronized video [19], whereas *VideoLLaMA2* combines spatio-temporal modeling and acoustic processing within a generative multimodal architecture [3]. More recent omnimodal architectures process multiple modalities within a unified framework. *Qwen2.5-Omni* adopts a thinker–talker architecture that accepts text, image, video, and audio inputs and produces textual or audio responses [14]. The subsequent *Qwen3.5-Omni* architecture integrates visual and audio encoders into a common backbone and introduces explicit temporal references to improve audio–video alignment [15]. These developments demonstrate that heterogeneous signals can be processed within a single architecture. However, the availability of a modality does not imply that it is used functionally or complementarily. This distinction between **multimodal availability** and **multimodal integration** is therefore essential. The former indicates that a system can accept multiple input sources, the latter requires the model to select, coordinate, and combine relevant information across modalities. A system may consequently be multimodal from an architectural perspective while relying predominantly on unimodal information during prediction. This issue is evident in *Audio-Visual Question Answering* benchmarks such as MUSIC-AVQA, which evaluates spatio-temporal relations in musical performance videos and extends the analysis to objects, entities, and everyday activities [12]. Subsequent studies reveal a strong visual bias, showing that many questions can be answered using the video stream alone, without exploiting the

[1] A multi-resolution Q-former, such as a Multi-Resolution Causal Q-Former, processes continuous audio, video, or image information at different temporal or spatial scales through distinct query banks, balancing fine-grained signals with broader contextual information.

audio channel. Under these conditions, similar audio-only and video-only performance does not demonstrate genuine integration, but rather suggests a form of unimodal collapse in which one modality dominates the decision process. To address this problem, Radevski, Gorjan, et al. (2026) introduced *Dave*, a benchmark in which each question jointly requires visual and acoustic information, making either modality insufficient in isolation [16]. The benchmark distinguishes among action recognition, sound classification, temporal ordering, and multimodal synchronization. This decomposition makes it possible to determine whether an error arises from the failure to perceive an event or from the inability to establish temporal relations between sounds and visual actions. The results show that models may achieve strong performance on individual components while encountering greater difficulty in audio–visual synchronization, indicating that cross-modal integration remains a central limitation. The increasing attention to audio has also led to benchmarks specifically designed to assess acoustic intelligence. *MMAU-Pro* evaluates 49 competencies distributed across speech, environmental sounds, music, and their combinations [11]. These tasks include long-audio comprehension, multi-track reasoning, spatial localization, prosody interpretation, and the integration of dialogue, music, and background noise. The persistently limited performance of current models indicates that acoustic processing cannot be reduced to dialogue recognition alone, since audio also conveys spatial, temporal, emotional, and contextual information that must be evaluated independently. A complementary perspective is provided by *SONIC-01*, a benchmark for omnimodal models based on audio–visual interactions drawn from real conversational and everyday contexts [17]. SONIC-01 combines open-ended summarization, multiple-choice questions, and temporal event localization, while also emphasizing that replacing raw audio with transcription removes paralinguistic information related to tone and communicative intention. Its results identify temporal localization as one of the most challenging tasks, confirming that the integration of verbal content, acoustic signals, and visual sequences remains an open problem. The comparison between native audio and transcription is also central to my recent work, *Lost in Translation: Investigating the Role of Audio Information for Video-based Visual Statement Verification* [5]. The study extends the MAIA framework by introducing audio into the VSV task and comparing two representations of the acoustic modality: raw audio directly processed by *Qwen2.5-Omni-7B* and automatic transcriptions supplied as additional textual input to *Qwen2.5-VL-7B*. To characterize the audio content, the videos are divided into four classes:

Music: Instances characterized by repetitive vocalizations, onomatopoeic patterns, strongly repetitive tokens, or explicit lexical cues associated with non-speech audio.

Dialogue: Videos whose transcriptions display sufficiently coherent linguistic content, acceptable log-probability values, low degeneracy, and limited evidence of no-speech segments.

Silence/Noise: Videos with empty or highly unreliable transcriptions, particularly when associated with low log-probability values and relatively high no-speech probabilities.

Unknown: Residual cases that appear speech-related but do not satisfy the dialogue criteria with sufficient confidence, thereby preserving potentially informative content while explicitly marking its uncertainty.

Class labels and speech transcriptions were obtained by combining the outputs of *Whisper-1* and *GPT-4o-transcribe*, together with an assessment of output plausibility. This classification makes it possible to distinguish results according to the informativeness of the available audio content. The results reveal a marked dominance of the visual channel. Audio provides only a limited benefit, mainly when no visual information is available, once rich visual input is provided, its contribution becomes negligible and may even introduce noise or errors, with limited robustness when a single black frame is added. Under visually rich conditions, the difference between with-audio and without-audio settings is therefore minimal, suggesting that audio functions more as a surrogate channel than as information fully integrated with vision. Audio transcriptions exhibit a different behavior. When added to already rich visual input, they reduce accuracy relative to the visual-only condition. This result shows that transcription is not a neutral transformation: once projected into the same representational space as the questions and answer options, it can compete with them, increasing the influence of textual regularities and introducing redundant or irrelevant information. *Lost in Translation* therefore shows that different representations of the same acoustic content can affect the model's decision process differently. Raw audio can be partially ignored when it is not useful, whereas transcription directly interacts with the linguistic component of the model. The introduction of an additional modality consequently does not guarantee an improvement and may instead compromise predictions that were already correct on the basis of reliable visual evidence. These findings must be interpreted in light of the original MAIA design. The benchmark was created to investigate visual content, and annotators were explicitly instructed not to consider audio information. Consequently, its questions and answers were not designed to exploit acoustic evidence. The limited contribution of audio may therefore reflect both the models' modality-integration mechanisms and the limited relevance of audio within the original dataset. Recent literature further emphasizes the importance of identifying the distinct sources of multimodal errors. *MIRAGE* disentangles hallucinations caused by perceptual failures from those arising from incorrect reasoning over correctly represented information [6]. Although primarily focused on visual reasoning, its methodological principle extends naturally to audio–visual settings, where an incorrect response may derive from failed sound recognition, inaccurate temporal synchronization, selection of the wrong modality, or erroneous inference over otherwise correctly represented evidence. Complementary attribution approaches seek to identify which parts of the input contribute to the generation of an answer. *OmniTrace* proposes an attribution framework for omnimodal autoregressive generation that links generated tokens to textual segments, visual regions, and audio/video intervals by aggregating attention signals into cross-modal explanations associated with linguistically interpretable units. This type of analysis is particularly relevant for determining whether a model actually exploits audio information or whether its prediction is driven primarily by text and vision. The current state of the art therefore reveals a clear tension between the rapid development of omnimodal architectures and their effective ability to integrate heterogeneous modalities. Contemporary models can receive multiple input types within the same context, yet diagnostic benchmarks provide no guarantee that these sources are used complementarily. Performance may instead depend on a dominant modality, linguistic shortcuts, dataset correlations, or additional signals that introduce noise. A central unresolved limitation is the construction of benchmarks in which audio and visual information are explicitly supervised. Assessing genuine audio–visual comprehension requires items in which audio contributes relevant information and the correct answer cannot be determined from a single modality alone. Such a design would make it possible to distinguish complementarity, dominance, substitution, and interference, while evaluating whether the model selects the appropriate evidence and combines it coherently across modalities. The present study follows this direction. Building on the competence-oriented MAIA framework and the recent findings of *Lost in Translation*, it investigates model behavior on a dataset specifically designed to require a genuine contribution from audio in order to answer the questions correctly. The central hypothesis is that, when audio and video provide partial and complementary evidence, multimodal comprehension depends on the model's ability to reconstruct the missing information through cross-modal integration, thereby avoiding both unimodal collapse and linguistic shortcuts based on plausibility.

2. Data Collection

2.1 The Original Video Dataset and Audio Enrichment

The benchmark **Multimodal AI Assessment - Audio Visual (MAIA-AV)**, created for this experiment, is an expansion of the original *Multimodal AI Assessment (MAIA)* dataset. The original dataset was developed as a native Italian collection of videos and tasks oriented to the fine-grained assessment of the reasoning and comprehension abilities of Visual Language Models (VLMs). The advantage of this design is that both the video content and linguistic data were natively created within an italian linguistic context. The original set of one hundred videos has an average duration of approximately thirty seconds per video. The dataset originates from *YouTube Italia*, the original selection guidelines emphasized italian cultural representation and thematic variety. Videos containing people and clearly observable actions were privileged to provide sufficiently rich visual evidence for formulating questions related to events, relationships, and temporal changes. The original *MAIA* structure was built around nine main reasoning categories and twelve fine-grained sub-categories:

Causal : Focuses on the causes and effects of events depicted in the video. It includes both explicit causal relations, which are directly observable, and implicit causes or effects that must be inferred from the available visual evidence.

Counterfactual : Concerns hypothetical events that do not occur in the video but could plausibly happen under different conditions. It evaluates the model's ability to reason about alternative scenarios while remaining grounded in the context of the video.

Implicit : Targets entities, actions, or events that are not directly visible but can be inferred from contextual information. The relevant information may be entirely absent from the video or may have appeared only during an earlier part of it.

Out-of-Scope : Introduces questions that presuppose the existence of entities or events that are not present in the video. It assesses whether the model recognises the absence of sufficient visual evidence instead of producing an unsupported or hallucinatory answer.

Planning : Focuses on the sequence of actions required to achieve a goal related to the situation shown in the video. It evaluates the model's ability to infer appropriate future steps from the available visual context.

Sentiment : Examines the emotions, attitudes, moods, or reactions expressed by the characters in the video. It tests the model's ability to identify affective states from observable visual cues and interactions.

Spatial : Concerns the position of entities and the spatial relationships between them. It includes both stable relationships that remain valid throughout the video and time-dependent configurations that apply only at specific moments.

Temporal : Focuses on the temporal organisation of the events shown in the video. It evaluates the model's ability to identify when events occur, determine their order, and estimate their duration.

Uncertainty : Addresses questions about entities or events that are visible but for which the video does not provide enough information to formulate a definitive answer. It evaluates the model's ability to recognise ambiguity and avoid unjustifiably assertive predictions.

For the purpose of **MAIA-AV**, we select only three of these categories, omitting their corresponding sub-categories: **Spatial**, **Temporal**, and **Causal**. This restriction is motivated by the need for a distinct performance evaluation based on question complexity, as detailed in Section 2.3.1.

2.2 Selection of Multimodal Materials

2.2.1 Video Collection Procedure

Per il dataset di MAIA-AV Sono stati selezionati sempre cento video, la dutata va dai 15 ai 45 secondi, sono tutti video in italiano che contengono dialoghi o narrazione delle scene che avvengono. La scelta di queste caratteristiche audiovisive non è casuale, ma è ben pensata sulla base delle indagini che si faranno dopo sul modello, c'è bisognio strettamente che nel video ci sia almeno un evento che abbia una causa e una

Technical walkthrough The first step of the dataset construction pipeline is executed via the `videodownloader.py`[4] script. The code is designed to retrieve video segments from *YouTube Italia* and save them in `.mp4` format. The core library utilized in the script is `yt_dlp`, which enables downloading audio-visual content from various online platforms, and also, it leverages the `download_range_func` function to specify the exact time interval to extract.

```
import yt_dlp
from yt_dlp.utils import download_range_func
```

When executed, the script prompts the user for the URL of the target video, the start time of the desired segment in `HH:MM:SS` format, and its duration in seconds. The start time is subsequently converted into a numerical timestamp in seconds using the custom `parse_time()` function:

```
def parse_time(time_str):
parts = [int(p) for p in time_str.split(':')]
return parts[0] * 3600 + parts[1] * 60 + parts[2]
```

The main function, `download_clip()`, handles user interaction and coordinates the extraction process. The script is configured to consistently retrieve the highest available quality format:

```
'format': 'bestvideo[height<=2160]+bestaudio/best'
```

The primary directive is utilized when *YouTube* streams audio and video separately. The fall-back option, `best`, serves as a secondary solution: whenever separate but compatible streams cannot be fetched, the function downloads the optimal pre-merged audio-visual stream. The time interval is defined via `download_range_func()`:

```
    'download_ranges': download_range_func(None,[(start_sec, end_sec)])
```

In this manner, `yt_dlp` instructs `FFmpeg` to download only the segment bounded by `start_sec` and `end_sec`. To prevent black or frozen frames, keyframe creation is explicitly forced at the cutting boundaries:

```
    'force_keyframes_at_cuts': True,
```

This process is necessary because compressed video formats do not contain full images in every frame, relying instead on inter-frame dependencies. Cutting at a position lacking a keyframe can result in corrupted frames at the beginning of the video segment. Forcing keyframes at cut points enhances the temporal precision of the segment, albeit at the cost of requiring content re-encoding. Finally, the output video is explicitly configured to the `MP4` format:

```
    'merge_output_format': 'mp4',
```

Following the segment download, `FFmpeg` re-encodes the video stream using the `H.264` codec and the audio stream using `AAC`:

```
    'postprocessor_args': {
        'ffmpeg': [
            '-c:v', 'libx264',
            '-preset', 'fast',
            '-c:a', 'aac'
        ]
    },
```

This step is fundamental to ensuring that every file in the dataset maintains a homogeneous configuration compatible with downstream processing tools. Such uniformity is required for subsequent frame extraction and `MP3` audio conversion.

2.2.2 Audio Conversion

2.2.3 Transcription Pipeline

2.3 Task Design and Collection

2.3.1 Questions Design

2.3.2 Answers Collection

3. Study Design and Experimental Setup

3.1 Preliminary Analysis as a Diagnostic Tool for Multimodal Assessment

The preliminary analysis aims to make the MAIA assessment more diagnostic and less dependent on final task outcomes. Performance on a Visual Question Answering (VSV) item alone does not reveal which prerequisite capabilities required by the task have been successfully satisfied. A correct answer does not necessarily imply that the model has accurately identified the relevant entities, reconstructed the events, established the required temporal relations, or correctly integrated the available evidence. Conversely, an incorrect answer does not indicate which of these components is responsible for the failure. This limitation is particularly relevant for MAIA, which evaluates fine-grained reasoning grounded in video content. For this reason, the preliminary analysis is performed independently of the VSV task. Rather than relying exclusively on direct question answering and aggregated accuracy, the video content is first represented through a structured analysis of entities, events, and relations among them. This representation is subsequently compared with an analogous analysis of the requirements imposed by the questions. Separating the two stages reduces the risk that video analysis is conditioned by the linguistic formulation of the question or by the plausibility of the available answer options. The objective is therefore to determine **what information can be retrieved from the video and what level of integration is required to retrieve it**, before relating this evidence to the demands of the VSV task. This framework has a methodological affinity with *VILMA* [10], particularly with its distinction between preliminary proficiency evaluation and the main test. *VILMA* implements this principle through a caption–foil paradigm in a zero-shot setting, using proficiency tests to verify prerequisite capabilities before interpreting performance on more complex phenomena. Our approach does not reproduce the *VILMA* evaluation format; rather, it adopts the underlying methodological principle that complex abilities should be interpreted in relation to the simpler perceptual, temporal, and semantic capabilities on which they depend. Accordingly, the proposed analysis distinguishes between tasks that require direct retrieval of localized information and tasks that require progressively greater integration across entities, events, temporal contexts, and inferential relations. In video understanding, this distinction is particularly important because relevant information may be contained within a single frame or distributed across a sequence. Questions belonging to the same nominal category may therefore impose substantially different processing requirements. A spatial question, for example, may require the retrieval of a location from a single frame or the reconstruction of a spatial configuration after a sequence of events. To capture this variation systematically, the classification is based on explicit operational criteria rather than on an intuitive notion of difficulty. The proposed question classification framework estimates the complexity of the operations required to retrieve and integrate the relevant evidence. A common three-level scale is applied to three dimensions: **spatial**, **temporal**, and **causal**. At Level 0, the required information can be retrieved directly without comparison or reconstruction. Level 1 requires the integration of an explicit relation or dependency involving a limited amount of evidence. Level 2 requires a broader reconstruction across events, temporal phases, state changes, or implicit relations. The same progression is applied consistently across the three dimensions, as summarized in Table 3.1. The classification criteria are grounded in properties of the evidence required to answer each question, including its temporal distribution, the number of relevant evidence units, the presence of event relations, and the explicit or implicit nature of the required information. These properties are intended to describe the informational requirements of the task rather than its linguistic formulation. An independent analysis of the video is therefore necessary to determine whether the entities, events, and relations presupposed by the question are actually recoverable from the available visual evidence. The video analysis is organized into successive stages of semantic representation. The preprocessing stage provides temporally indexed video content, including a dense representation used to preserve temporal information. Four omnimodal and multimodal models, subsequently evaluated on the VSV task, independently analyze temporally localized portions of each video. Their outputs are used to construct structured representations of the entities, events, and relations identified in the observed content. The analysis is performed independently of the questions so that the resulting representation reflects information retrieved from the video rather than evidence selected in response to a specific query. Temporally overlapping portions of the video are used to preserve the connection between extracted information and the intervals in which it is supported. This temporal anchoring is necessary for the subsequent comparison between the evidence represented by the models and the evidence required by individual questions. For each video, the analysis progressively considers three levels of representation: **entities**, **events**, and **relations between events**, including inferential relations where required. Outputs from the different models are initially kept separate, allowing inter-model agreement to be measured without prematurely merging their predictions. Agreement does not constitute ground truth, since different models may produce the same unsupported interpretation; it is instead treated as an indicator of **prediction stability**. The analysis therefore distinguishes information that is directly supported by observable evidence from information introduced through model inference. Within this framework, **event extraction** is treated as the identification of semantically structured events associated with specific temporal intervals. Because the same event may appear in multiple overlapping portions of the input, subsequent consolidation reduces redundancy while preserving genuinely repeated events. The consolidated representation can then be used to establish temporal relations such as *before*, *after*, *during*, and *overlap*, producing a temporally organized representation of the video. This distinction between entities and events is essential for evaluating temporal complexity. Knowing that a video contains a person, a mug, and a table provides a different type of information from knowing that the person picks up the mug, drinks from it, places it on the table, and subsequently walks away. The former represents the entities present in the scene, whereas the latter requires the representation of actions, temporal order, and changes of state. Similarly, identifying the location of an object at a given moment imposes different requirements from identifying its location after a particular event. The preliminary analysis is designed to make these differences explicit. The causal dimension introduces an additional inferential requirement. Temporal succession alone does not establish causality; consequently, causal relations must be distinguished from simple co-occurrence or temporal ordering. The analysis first represents the relevant events and their temporal organization. It can then be determined whether answering a causal question requires information that is explicitly available in the video or an additional inference connecting observed events, states, or intentions. A further objective of the preliminary analysis is to determine which information can be recovered from the visual modality independently of other channels. This addresses a methodological issue highlighted by previous experiments [5], in which visual information was compared with additional modalities such as audio and automatic transcription. In the original MAIA setting, the addition of these channels did not produce a systematic improvement, while the inclusion of audio transcriptions resulted in a small but statistically robust performance decrease. One relevant characteristic of the original dataset is that its VSV items were constructed around visual content rather than around information specifically provided by the acoustic channel. For this reason, the preliminary analysis establishes a visual baseline focused on *what can be seen* before assessing the contribution of *what can be heard*. This separation makes it possible to evaluate whether acoustic information provides genuinely complementary evidence and whether questions designed for multimodal assessment require information distributed across more than one modality. The purpose of the preliminary analysis is therefore to characterize **which visual information is available in the video and how much evidence must be retrieved and integrated to answer a question**. This representation provides the basis for evaluating whether the models subsequently tested on the VSV task are able to recover and combine the information required by each item. In this regard, the preliminary analysis moves *MAIA-AV* beyond a purely result-based evaluation towards a more diagnostic and competence-oriented framework. Building on the methodological principle underlying the proficiency tests introduced by *VILMA*, the proposed approach makes the prerequisite conditions for successful question answering explicit through a structured analysis of video content. This allows model performance to be interpreted in relation to specific task requirements, distinguishing limitations in entity recognition, event identification, temporal integration, spatial reasoning, and causal inference rather than attributing an incorrect response generically to a failure of multimodal reasoning.

Category	Level 0	Level 1	Level 2
Spatial (S)	Direct retrieval of an entity's location or spatial property from a single frame.	Explicit spatial relation or event-conditioned localization involving at most two evidence units.	Spatial reconstruction across multiple phases, including movement, relocation, or state change.
Temporal (T)	Recognition of a single event or state without temporal comparison.	Temporal localization or pairwise ordering between two explicit events.	Multi-event temporal reconstruction, including duration, repetition, onset, or state evolution.
Causal (C)	Retrieval of an explicitly stated cause without inference.	Explicit cause–effect relation involving at most two evidence units.	Implicit or multi-event causal reconstruction requiring contextual or intentional inference.

Table 3.1: Complexity levels for spatial, temporal, and causal questions.

4. Results and Discussions

5. Conclusions

Conclusioni...

6. Future Perspectives

List of Abbreviations

MAIA-AV	Multimodal AI Assessment - Audio Visual
VLM	Visual Language Model
VQA	Visual Question Answering
AGQA	Action Genome Question Answering
MVBench	Multi-modal Video Understanding Benchmark
VILMA	Video Language Model Assessment
MAIA	Multimodal AI Assessment
VSV	Visual Statement Verification
OEVQA	Open-Ended Visual Question Answering
MLM	Multimodal Language Model
AVQA	Audio-Visual Question Answering
Dave	Diagnostic benchmark for Audio Visual Evaluation
MMAU-Pro	Massive Multi-Task Audio Understanding and Reasoning benchmark, Pro version
SONIC-01	SOcial Natural Interaction Corpus (Omnimodal, v1)
GPT	Generative Pre-trained Transformer
MIRAGE	Assessing Hallucination in Multimodal Reasoning Chains of Multimodal Large Language Models

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A. Appendice